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PROCESS MINING AND PRODUCTION ROUTING FAST PROFILING FOR DATA-DRIVEN DIGITAL TWINS

Paulo Victor Lopes¹, Giovanni Lugaresi², Filipe Alves Neto Verri³, and Anders Skoogh⁴

¹ Department of Industrial and Materials Sci., Chalmers University of Tech.: paulov@chalmers.se

¹ Department of Computer Methods, Instituto Tecnológico de Aeronáutica: paulo.lopes@ga.ita.br

² Department of Mechanical Engineering, KU Leuven: giovanni.lugaresi@kuleuven.be

³ Department of Computer Methods, Instituto Tecnológico de Aeronáutica: filipe.verri@gp.ita.br

⁴ Department of Industrial and Materials Sci., Chalmers University of Tech.: anders.skoogh@chalmers.se

Abstract—Industry increasingly focuses on Digital Shadows and Twins of production lines, especially for planning, controlling, and optimizing operations. In parallel, shop floor processes can be described using Discrete Event Simulation (DES) models, which are ranked among the top tools for manufacturing system decision support. Although, Process Mining (PM) and model-driven Digital Twins (DT) were investigated in separate research communities. The integration of these two research fields is essential for advancing industrial applications by reducing time and efforts to model and describe processes. Thus, the objective of this paper is to propose a data integration pipeline to enhance realistic event logs and support the early stages of Data-driven Modelling of DT through PM techniques. This paper is expected to provide three relevant contributions. The first contribution is the enhancement of the production system event logs through the implementation of data integration techniques. The second contribution is to enable machine learning techniques to be applied by trace profiling the enhanced event logs, generating an attribute-value database. The third contribution is to extract value from a process-centered analysis, increasing the data value from a practical perspective.

I. INTRODUCTION

Industry increasingly focuses on Digital Shadows and Twins of production lines, especially for planning, controlling, and optimizing operations [1]. In parallel, shop floor processes can be described using Discrete Event Simulation (DES) models, ranked among the top three tools for manufacturing system decision support [2]. DES can be included as a digital model for a Digital Twin development in a manufacturing system context [3]. However, accurate process models are often not available for data-driven performance analysis [4].

Process Mining (PM) algorithms are becoming increasingly important to facilitate data-driven insights, construction, and update of models in production systems[5]. However, process mining and model-driven digital twins were investigated in separate research communities [6]. This integration is essential for advancing this field of research, justified by the fundamental need to develop methodologies for discovering and analyzing industrial processes.

Therefore, process mining techniques are used to support or discover the simulation model from event

logs allowing structured creation of a first simulation model (to be evaluated and possibly modified). Rozinat first applied this strategy and had the objective of extracting parameters that could be used to create a simulation model [7]. However, it is imperative to have a good understanding of the processes and available databases to draw conclusions. Domain and modeling expertise are not rendered obsolete by the discovering techniques. This approach should be seen as a tool to support insights generation [8].

PM in Industry 4.0 still faces a relevant gap related to the lack of new PM technologies implementation in realistic working environments [9]. The authors argue that the first challenge faced is the pre-processing of realistic event logs to generate useful insights for Process Mining implementation in the Industry. Thus, this paper's objective is to propose a data integration workflow to enhance realistic event logs and support the early stages of Data-driven Modelling of Digital Twins and PM techniques. To achieve this general objective, the following specific objectives are proposed:

- 1) Propose a pipeline to integrate event logs and machine state databases;
- 2) Apply trace profiling techniques in the integrated dataset;
- 3) Use Process Mining techniques to gain insights from the trace profile database.

This paper is expected to provide three relevant contributions. The first contribution is the enhancement of the production system event logs through the implementation of data integration techniques. The second contribution is to enable machine learning (ML) and artificial intelligence (AI) techniques to be applied by trace profiling the enhanced event logs, generating an attribute-value database. The third contribution is to extract value from a process-centered analysis, increasing the data value from a practical perspective.

II. CONTEXT AND MOTIVATION

Recently, PM techniques increased in use for the analysis of productive systems and the consequent automatic simulation model generation. PM techniques have been used for the discovery and validation of process models in industrial applications [9]. As well as

discovering and recognizing production policies [10], modeling product arrival times [11] and analyzing the causes of process delays [12].

Several relevant information can be extracted as a result of the processes of production events detection and labeling. Event logs are produced and further used for process discovery using process mining algorithms [13]. Both processes and events that have been identified can be used to build data-driven simulation models for a given smart factory [14]. Once an extracted model is validated, it can be utilized for performing what-if scenarios, which are then evaluated according to predefined performance indicators. The results of these experiments can, finally, support the decision-making of stakeholders.

A. Industrial Process Mining and Data-Driven Digital Twins

Efforts focused on retrieving daily availability records from an event log [15] and how operational activities are batched [16]. Thus, theoretical works were carried out, such as the link of colored Petri Net states and exceptional system states [17] and the benchmark of process mining algorithms' ability to discover a digital model of a job shop [18]. In the past few years, research has started to evolve toward the elaboration of more complex frameworks [19].

Discovered simulation models can be used to improve certain manufacturing activities parameters [20]. Then, the integration of discovering the material flow, estimation of simulation parameters, and identification of control policies [21]. At this stage, the initial concepts enabling data driven digital twins emerged. Some examples of data-driven modeling of Digital Twins are given in the next paragraph.

Data driven frameworks were proposed for remanufacturing simulation model generation [22], for event graph modeling [23] and a knowledge-driven Digital Twin was developed [24]. The development of a method for generating simulation models with a proper level of detail was validated in a lab scale Digital Twin environment [25], [14]. In the current state-of-the-art process mining combined with machine learning techniques are applied to build a digital twin in a smart factory real case context [26]. Also, DT can be generated through process mining techniques extracting the main process map from a complex system, enabling production systems analysis as bottleneck detection, diagnosis, or improvement [27].

In this sense, there is a clear movement toward the utilization of process mining as a basis for the development of Data-Driven Digital Twins and industrial applications. The authors argue that a natural step for the improvement of these approaches is the enhancement of event logs, generation of what-if scenarios, and the use of machine learning/artificial intelligence techniques. To do so, trace profiling techniques offer a solution to transform event logs to an attribute value format, facilitating further work in this research field.

B. Trace Profiling for Process Mining

Trace profiling is a data-driven approach that involves examining traces to gather insights into the behavior, patterns, and variations within the process. It is based on the idea of generating homogeneous subsets of traces that can each be described by a proper process model [28]. The general idea is to use a means of grouping traces so to represent a particular behavior of the system for each group. Naturally, clustering techniques are among the most used for this scope, be that on traditional event logs, or object-centric [29]. Association rules to study the common behavior evident in certain profiles are useful to classify and generate useful knowledge (for instance, longer lead times depending on a specific faulty resource). Trace profiling has been applied in several applications spanning from profiling user interactions in information systems [30], cybersecurity [31], smart manufacturing [32], as well as Education [33].

In the context of manufacturing, trace profiling involves the collection and analysis of data traces that capture the sequence of events and interactions within the production processes. For instance, Maeyens et al. [32] analyzed trace profiles coming from CNC machines in comparison with traces generated with nominal behavior to identify abnormal patterns, using the comparison to validate the structure of the model, variations in the trace durations, as well as the frequency of relative paths.

Trace profiling is a promising technique to identify clusters of part types characterized by a different behavior. Part routing is of high interest to production planners as different routes can be designed based on specific characteristics of products, as well as on quality-based scores. It is therefore necessary to use available data to characterize the nominal part routing behavior before generating alternatives. To the best of authors knowledge, trace profiling has not been used to identify part-type specific routing behavior at system level.

III. METHODOLOGY

Usually, the event log and machine state databases are processed separately to calculate specific parameters for simulation models or data analysis [14], [26], [27]. This paper proposes to extract more insights from these databases through proper data integration, description, and profiling techniques. Our methodology starts with raw data consisting of event logs and machine state data, further integrated using timestamp information.

The output of the data integration strategy is the event log data in tabular format, containing the product ID, machine ID, timestamp, and machine state duration. At this point, the database is described through statistical analysis. A trace profile is then applied to structure these events effectively and enable machine learning application. This methodology step enables a deeper understanding of the data but also paves the way for process understanding through machine learning and artificial intelligence [28].

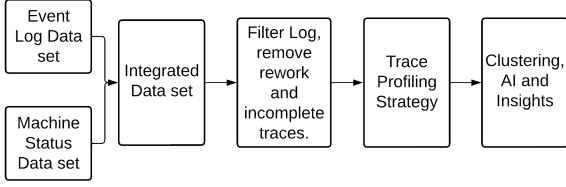


Fig. 1: Graphical workflow proposed by methodology.

We developed two routines to analyze the event log and the trace profile. The first routine, `analyze_log`, performs a descriptive analysis of a filtered event log. The second routine, `trace_profile`, creates a trace profile from a given event log. The trace profiling technique includes the creation of a dataset model with two basic information: the unique machine identification and the machine state duration. Finally, a clustering technique was applied to verify the feasibility of trace profiling as an input for machine learning techniques, the clusters were compared through T-test.

A. Use Case - Data Set Description

The data used consists of the machine state and event log data from a real-world production system from the Swedish Automotive Industry. In this case, the automotive parts production system is composed of five workstations connected by conveyor belts, and two different parts models. During a production run, in each workstation, an operator records the log when a part arrives and the state of the machine.

Machine_Id	Product_Id	Time_stamp
OP20-1	C003	2009-11-02 07:47:52
OP30-1	C002	2009-11-02 07:58:40
OP20-1	C004	2009-11-02 07:59:10
OP40-1	C024	2009-11-02 08:03:25
OP30-2	C001	2009-11-02 08:05:59
OP40-2	C022	2009-11-02 08:06:42
...	...	...

TABLE I: Sample of real event log from the shop floor.

The database that contains information on parts identification, production steps and the time stamps of parts arrival are described as event logs. Information about the part arrival and machine state is recorded with the corresponding timestamp, as in Table I. It is worth noticing that this event log do not include the end log of each activity, as many other logs from real shop floors.

Machine_Id	state	Code	Duration	Time_stamp
OP20-1	Down	0	05:12.0	07:15:16
OP40-1	Down	0	06:05.0	07:19:11
OP20-1	Down	0	02:20.0	07:20:28
OP40-2	Idle	3	02:54.0	07:21:21
OP30-1	Idle	3	01:27.0	07:21:22
OP30-2	Idle	3	04:49.0	07:21:23
OP20-1	Down	0	00:10.0	07:22:48
...	...	...	...	...

TABLE II: Sample of real machine state database from the shop floor.

State database stores the recording of the shop floor assets state. Which means the information about idle time, producing time and downtime is extracted from the machine state database, as Table II. This can be considered a low-level source of information with reduced efforts to collect [34].

B. Analyze log routine

The routine `analyze_log` performs a descriptive analysis of a filtered event log (Details about the filtering process are presented in Section IV-A). The analysis includes the calculation of the number of variants, the number of traces, the average time elapsed between cases, the histogram of case durations, and other statistics of case durations. Finishing with a Petri Net representing the discovered model from the event log.

A variant is a unique sequence of activities that occurs in the trace. As it is a relevant information, the routine displays the number of variants and their frequency. It also calculates the average time elapsed between the start of new cases. This is a measure of the average time between the start of a case and the start of the next case. This measure is useful to understand the case arrival rate, a basic parameter for simulations. The case duration is the time between the start and the end of a case. It is calculated for all cases in the event log, later displaying a histogram. This histogram is useful to understand the distribution of case duration's and identify outliers.

Finally, the heuristic miner algorithm is applied to the event log and visualizes the Petri net. The heuristic miner is a process discovery algorithm that infers a Petri net from an event log. Since the heuristic miner algorithm is parameterized by a frequency threshold, this method enables the user to choose the most appropriate value. Strictly speaking, one can choose any process discovery algorithm available in the PM4PY library [35]. In this implementation, we use the heuristic miner, as it generates simple models with good generalization and fitness for short and non-dense productive lines.

C. Trace profile routine

The `trace_profile` routine creates a trace profile from a given event log. The trace profiling technique includes the creation of a dataset model with two basic information. First, the unique machine identification (OP020-1, OP030-1, ...) and, second, the machine state duration (Time Idle, Time down, ...).

After the basic dataset model is created, the routine assigns the values for each row. This step is performed for each trace inside the log. Then, it is checked if the unique machine identifiers are part of this trace. If the machine is part of this trace, the machine state values are assigned to this row, otherwise, not a number is assigned.

The rows of the trace profile data set (pandas data frame) are identified by the trace identification. Which means that each row is identified by the product identifier. The outputs for this routine are: a) a data

frame in the format of attribute value representing the original event log information, b) a box plot that shows the distribution of machine state for each machine.

IV. RESULTS

The integration of event logs and machine state database was made through time stamps, resulting in the sample Table III where all observations are from 2009-11-02. The sum of state durations was assigned to each product between the time it arrived and when a new product arrived at the same machine workstation. The inability to validate the exact moment that a product leaves the workstations is known as a limitation of this dataset, as it is a practical case without logs of events end.

M_id	P_id	Time_stp	Time_P	Time_I	Time_D
OP20_1	C001	07:13:26	0:22:53	0:00:47	0:00:13
OP20_1	C002	07:37:18	0:09:45	0:00:47	0:00:02
OP20_1	C003	07:47:52	0:09:45	0:01:31	0:00:02
OP20_1	C004	07:59:10	0:09:46	0:00:47	0:00:01
OP20_1	C005	08:09:44	0:09:46	0:00:45	0:00:02
OP20_1	C006	08:20:17	0:09:55	0:00:27	1:27:28
OP20_1	C007	09:58:01	0:10:01	0:00:48	0:15:23
OP20_1	C008	10:24:18	0:12:12	0:00:45	0:00:01
OP20_1	C009	10:37:19	0:09:48	0:00:00	0:00:47
OP20_1	C010	10:47:53	0:09:47	0:00:00	0:00:46
...	...	...	...	...	...

TABLE III: Sample of the real integrated database from the shop floor.

It is possible to observe a lack of information about activities' end time stamps. Once the event log data is recorded, the only insight extracted in practice is the number of parts that a machine produces. Some consequences of the lack of relevant information are the difficulty in calculating and validating the activity durations, waiting times or processing times. Ultimately, these parameters can become impossible to validate.

A. Analyze log routine

The initial integrated dataset has incomplete traces and instances of rework, presented in Figure 2. The process model representing this event log is discovered through the application of a process discovery technique called the heuristic miner algorithm. It becomes necessary to pre-process this dataset to get useful information.

The pre-processing phase includes the removal of anomalies, such as the selection of traces starting at OP020-1, finishing at OP040-1 or OP040-2, excluding traces with rework. The outcome is a cleansed process that represents the system behavior without noise, presented in Figure 3. This filter enhances the quality of subsequent analyses but the pre-processing step also contributes to a more informed description of business operations.

Worth noticing that in Figure 2 there were 343 traces, and this number was reduced to 216 traces in Figure 3. In Figure 3, the Petri Net model shows a clear distinction between the first, second, and third

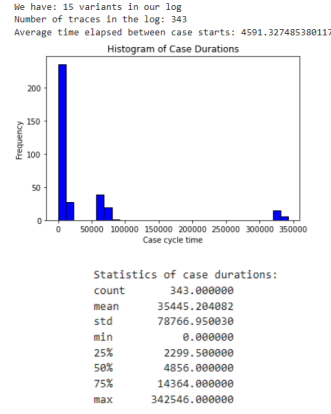


Fig. 2: Directed followed graph result from the original event log dataset.

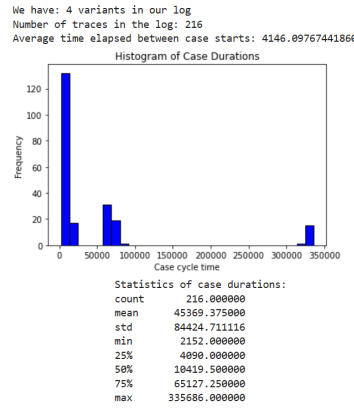


Fig. 3: Directed followed graph result from the pre-processed event log dataset.

production steps. Also, the behavior of the case duration histogram remains the same, with 3 groups of values. These 3 groups represent different states of the productive system.

B. Trace profile routine

The output of trace profile routine is an extended data frame in the format of IV, including three columns like OP020-1 description for all the other machines. This information is displayed on the data frame rows, for all the products of the pre processed event log data set, constituting the machine state profile. The columns represents the Time Processing (TP) for each product, Time Idle (TI) when the machine is blocked or starved, and Time Down (TD) when the machine is turned off. As the end event of each process is not recorded in the log, the product process time ends when a new product enters the machine.

P_id	OP020_1_TP	OP020_1_TI	OP020_1_TD	...
091102C001	1373.0	47.0	13.0	...
091102C004	586.0	47.0	1.0	...
091103C007	598.0	43.0	2.0	...
091103C008	577.0	2.0	44.0	...
...	...	...	...	...

TABLE IV: Trace Profile example presenting values in seconds.

Figure 4 presents the histogram of two different strategies to calculate the case duration of traces. Strategy 1 (S1), in Figure 4(a), calculates the case durations by PM4PY, the same usually utilized in Python extensions to simulate Petri Nets [36]. Strategy 2 (S2), in Figure 4(b), calculates the case durations through the integrated data set from the machine state profile.

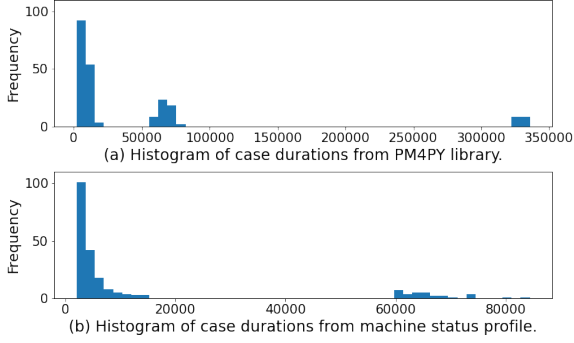


Fig. 4: Comparison of case durations obtained through trace profiles and PM4PY.

The histograms in Figure 4 show a difference in the distribution of values. The authors considered that S1 includes the waiting times after the product leaves the machine. While S2 considers the production process from a machine perspective. This analysis shows that S2 provides more detail about parameters that are inputs for simulation, such as producing time, idle time, and downtime. The waiting times are usually an output of a discrete event simulation (DES) model, as a data-driven discrete event simulation digital twin (D3ES-DT). It is possible to observe that the use of machine status profiles shows evidence to be a contribution to improving the accuracy of D3ES-DT models. This also applies to other industrial applications, such as bottleneck analysis [37].

C. Machine learning application as proof-of-concept

The trace profiles were standardized before cluster. KNN was used with 2 clusters, and silhouette was calculated but increasing values until 42 clusters, so we decided to use just 2 clusters to validate and analyze. The clustering techniques were applied to traces with a case duration calculated from a machine state profile of less than 40000 seconds, representing the group of Figure 4(b). This first group represented the traces without a shift change, represented by the group in between 60000 and 80000 seconds of case duration.

Figure 5 shows a box plot of trace clusters without outliers. The outliers were removed to facilitate the visual analysis. However, this also means that some observations of OP040_2 and OP030_2 were hidden from Figure 5(b) and observations of OP030_1 were hidden from Figure 5(a). Table V was then built to statistically describe the case durations from machine state trace profiles and explore these results.

According to Table V Cluster 1 has a fewer number of observations compared to Cluster 2, determined

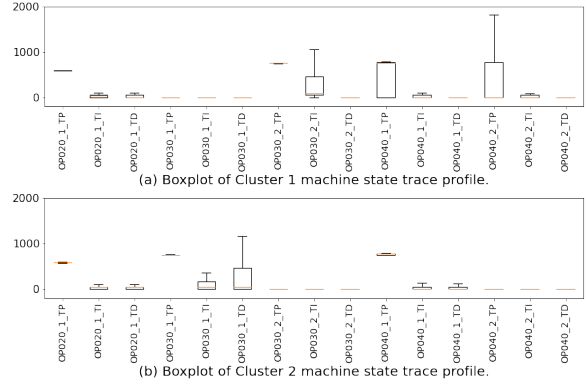


Fig. 5: Description of trace profile on two different clusters of production, without outliers.

Cluster	count	mean	std	min	max
Cluster 1	66	4883.1	3156.6	2255	15114
Cluster 2	117	4181.3	2295.6	2128	13332

TABLE V: Statistical description of clusters of production traces.

by the processing capacity of OP030-1 and OP030-2. In Figure 5 it is possible to verify the existence of two main routes for the products after the second production step. These routes are determined by the product traces and production parameters. Cluster 2 also represents the more stable route, as it is more frequent, lower standard deviation, and lower mean case duration. Cluster 1 represents the secondary route more variable in terms of topology and case duration.

Hypothesis H_0 (Null hypothesis): The mean case duration of Cluster 1 is less than or equal to Cluster 2 mean case duration.

Hypothesis H_1 : The mean case duration of Cluster 1 is greater than Cluster 2 mean case duration.

Hypotheses H_0 and H_1 are tested according to T-test for the means of two unpaired independent samples of scores with a significance level of 0.05 (p-value) using the SciPy package in Python [38]. The obtained T-statistic was 1.7348 and the p-value of 0.0422, which indicates that the null hypothesis can be rejected. However, the authors argue that for further analysis the clusters should be revised in order to achieve p-values lower than 0.02 for a highly significant statistical statement.

D. Further works

Subsequently, we propose to improve the cluster analysis to discover different product routes and what-if scenarios. In the further steps of D3ES-DT, the transition from process models to simulation models can use a SimPy-based strategy tailored for simulating Petri nets [36]. This strategy is designed to replicate the real-world production scenarios efficiently adapting fast to new scenarios. Essentially, a digital twin of a production system can be created from event log data, offering a powerful tool for scenario analysis, optimization and decision-making.

V. CONCLUSIONS

Process Mining techniques offered valuable insights for the processing phase and process characterization. The integration of event log and state machine data can increase the amount of useful information for process analysis. Also, real data sets with missing observations can benefit from this extra information when modeling systems. Trace profiles enable the easy application of ML/AI algorithms.

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AUTHOR BIOGRAPHIES

PAULO VICTOR LOPES is a Research Assistant at the Industrial and Material Science Department of Chalmers University of Technology and Ph.D. student in Operations Research at Instituto Tecnológico de Aeronáutica and Universidade Federal de São Paulo. His research interests include digital twins, traceability systems, process mining, digital platforms, data-driven decision making and production systems. His e-mail address is paulov@chalmers.se.

GIOVANNI LUGARESI is Assistant Professor at the Department of Mechanical Engineering of KU Leuven. His research interests include production planning and control, internal logistics, process mining, and robust optimization. His email address is giovanni.lugaresi@kuleuven.be.

FILIFE VERRI is currently an Assistant Professor with the Division of Computer Science, Department of Computer Methods, Instituto Tecnológico de Aeronáutica (ITA). His research interests include data science, machine learning, complex networks, and complex systems. He is also a founding member of the DroneComp Research Group, that focuses on developing computer systems and computational methods for the next-generation air transportation system. Before starting his Professorship career, he graduated summa cum laude in bachelor and Ph.D. from the University of São Paulo. His email address is verri@ita.br.

ANDERS SKOOGH is a Professor at Industrial and Material Science, Chalmers University of Technology. He is a research group leader for Production Service & Maintenance Systems. Anders is also the director of Chalmers’ Masters’ program in Production Engineering and board member of the think-tank Sustainability Circle. Before starting his research career, he accumulated industrial experience from being a logistics developer at Volvo Cars. His email address is anders.skoogh@chalmers.se.